

Agenda:

1. Evolution of containers
2. What is Docker
3. Why do we need Docker
4. Docker Lifecycle and Terminologies
5. Docker Use-cases
6. Docker Setup, Verify Installation
7. Docker Hub - Push & Pull the Images
8. Dockerize the FastAPI application.
9. Docker Compose - Intro & Hands-on

Why do we need Containers if we had VMs?

- No separate OS.
- VMs will be having separate OS

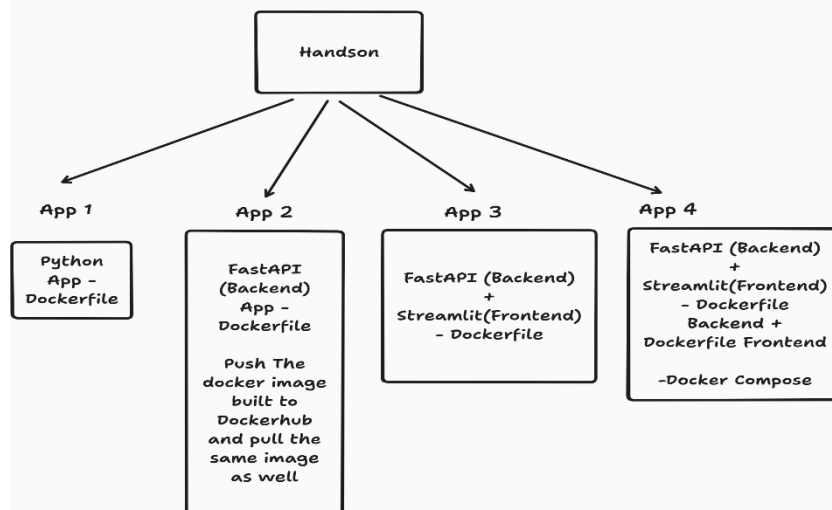
Docker: Docker is a platform designed to help developers build, share and run container applications.

Analogy:

Why do we need Docker:

1. Consistency across Env
2. Isolation
3. Scalability

Build, Ship, Run



Steps to be followed for creating Docker Image/Container: (APP 1)

1. Create a simple python application in py file format - main.py
2. Create a requirements.txt file
3. Dockerfile and add instructions to build the Image.
4. Create Docker Image using build command
5. Running the Docker Images to create a Container.

Docker Container

app:
requirements.txt

1. docker build (Create Image)
2. docker run (Run the container from the image created)